

# CSIS1624 2D Array

DATE: 22 July 2026

- Make sure to include a comment block with your details at the very top of your file. Marks will be subtracted for every missing or incorrect occurrence.
- Your program should run. Marks are awarded for an executable program.
- Code that is commented out will not be marked.
- You must use comments to explain your program logic and mark the different sections of your code. Comments should be meaningful. Marks will be subtracted for failing to include comments.
- Make sure that your code is neat i.e. no misaligned braces or overuse of empty space. Marks may be deducted for untidy code.
- You may **not** use any form of Generative AI or any other online resource at any point. This will result in an automatic 0 (zero) mark. You may only use the material on Blackboard.
- **READ YOUR INSTRUCTIONS CAREFULLY. Read the entire document before attempting the activity.**

## Scenario / Problem Specification

Develop a program that works with a **two-dimensional (2D) integer array**. The program should ask the user to enter the **number of rows** and the **number of columns** for the matrix.

The program should then prompt the user to fill each cell of the matrix using nested for loops. Each cell is identified by its row and column index.

Once all values have been entered, the program will display the matrix on the console — one row per line — so that the values are laid out in a grid format.

## Code the Solution

1. Create a Console (.NET framework) App using Visual Studio 2019 named **MatrixInput** on the H-drive. Remember the location where you saved the files.
2. Ask the user to enter the **number of rows** and the **number of columns** for the matrix. Use **int.Parse** to read each value. (see **Figure 1**)
3. Declare a 2D integer array named **matrix** using the rows and columns entered by the user.
4. Using **nested for loops**, prompt the user to enter a value for each cell of the matrix. Display the current row and column index in the prompt (e.g. **matrix[0,1] =** ). (see **Figure 2**)  
*Hint: use **Console.Write("matrix[{0},{1}] = ", row, col)** inside the inner loop to display the formatted prompt.*
5. Using a second set of **nested for loops**, display the matrix on the console. Each row should be printed on a new line. Use **matrix.GetLength(0)** and **matrix.GetLength(1)** to determine the bounds. (see **Figure 3**)

```
Enter the number of the rows: 3
Enter the number of the columns: 3
```

Figure 1 User input for matrix dimensions

```
Enter the cells of the matrix:
matrix[0,0] = 1
matrix[0,1] = 2
matrix[0,2] = 3
```

```
matrix[1,0] = 4
matrix[1,1] = 5
matrix[1,2] = 6
matrix[2,0] = 7
matrix[2,1] = 8
matrix[2,2] = 9
```

Figure 2 Cell-by-cell input using nested for loops (3x3 example)

```
1 2 3
4 5 6
7 8 9
```

Figure 3 Matrix displayed as a grid (program output)

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### Submitting a Visual Studio solution folder

Submit your **zipped** Visual Studio solution folder on Blackboard. You may refer to the submission process document on Blackboard (CSIS1614 Submission Process 2026).

- Make sure to name the zipped folder as **Surname (StudentNumber)** e.g. Doe (2025123456)
- Marks are allocated for a running program and all the files in the Visual Studio solution folder are necessary to run your program.

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### Submission checklist

- ✓ I have added a comment block with my details at the top before the using directives.
- ✓ I have zipped my entire solution folder and named it correctly: Surname (StudentNumber).
- ✓ I have downloaded my zipped folder to check that I have submitted the correct files (You must submit the entire project folder, not just the .sln file).